

Algorithms

Lecture # 01

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Design and analysis of Algorithms

Design and Analysis of Algorithms
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Model of Computation

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- In order to say anything meaningful about our algorithms, it will be important for us to settle on a mathematical model of computation
- Ideally this model should be a reasonable abstraction of a standard generic single-processor machine.

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- Ideally this model should be a reasonable abstraction of a standard generic single-processor machine.
- We call this model a *random access machine* or *RAM*.

Preliminaries

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- *Course:* Design and Analysis of Algorithms
- *Prerequisites:* Data Structures, Discrete Mathematics
- *Textbook:* Introduction to Algorithms, Cormen, Leiserson, Rivest, Stein, 2nd edition, 2001 The MIT Press.

Preliminaries

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- *Textbook:* Introduction to Algorithms, Cormen, Leiserson, Rivest, Stein, 2nd edition, 2001 The MIT Press.
- Grading:
 - homework and projects
 - mid term exam
 - final

Origin of Word: ALGORITHM

Origin of word: Algorithm

Abu Ja'far Mohammad
ibn Musa al-Khowarizmi.

Origin of Word: ALGORITHM



Origin of Word: ALGORITHM



Origin of Word: ALGORITHM

Origin of word: ***Algorithm***

- His year of birth is not known exactly.
- Al-Khwarizmi parents migrated to a placesouth of Baghdad when he was a child.

Origin of Word: ALGORITHM

Origin of word: ***Algorithm***

- It has been established from his contributions that he flourished under Khalifah Al-Mamun at Baghdad during 813 to 833 C.E.
- Al-Khwarizmi died around 840 C.E.

Origin of Word: ALGORITHM



Origin of Word: ALGORITHM

*al Kitab al-mukhatasar
fi hisab al-jabr wa'l-
muqabalah*

Origin of Word: ALGORITHM

Origin of word: ***Algorithm***

- It is from the titles of these writings and his name that the words **algebra** and **algorithm** are derived.
- As a result of his work, al-Khwarizmi is regarded as the most outstanding mathematician of his time

ALGORITHM informal definition

Algorithm: Informal Definition

- An algorithm is any well-defined computational procedure that takes some values, or set of values, as input and produces some value, or set of values, as output.
- An algorithm is thus a sequence of computational steps that transform the input into output.

ALGORITHM, programming

Algorithms, Programming

- A good understanding of algorithms is essential for *programming*.
- However, unlike a program, an algorithm is a *mathematical entity*

ALGORITHM, programming

Algorithms, Programming

- Algorithm is independent of a specific programming language, machine, or compiler.
- Algorithm design is about mathematical theory behind the design of good programs.

Why Study?

Why Study Algorithms

- This course is all about how to design good algorithms.
- **CS301** dealt with how to design good data structures.
- This is not really an independent issue because the fastest algorithms are fast because they use fast data structures, and vice versa.

Why Study?

Why Study Algorithms

- The fact is that many of the courses in computer science deal with efficient algorithms and data structures.

Why Study?

Why Study Algorithms

They apply these to various applications:

- compilers,
- operating systems,
- databases,
- artificial intelligence,
- computer graphics and vision,
- networks, etc.

Why Study?

Why Study Algorithms

- A good understanding of algorithm design is a *central element* to a good understanding of computer science and computer programming.

Course outline

Course Outline

This course will consist of four major sections.

1. Mathematical tools necessary for the analysis of algorithms. This will focus on asymptotics, summations, recurrences.

Course outline

Course Outline

2. Sorting: this will focus on different strategies for sorting, and use this problem as a case-study in different techniques for designing and analyzing algorithms.

Course outline

Course Outline

3. Collection of various algorithmic problems and solution techniques: Dynamic programming, greedy strategy, graphs.

Course outline

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4. Introduction to the theory of NP-completeness.
NP-complete problem are those for which no efficient algorithms are known, but no one knows for sure whether efficient solutions might exist.

Criteria for analyzing Algorithms

Criterion for Analyzing Algorithms

- In order to design good algorithms, we must first agree on the *criterion* for measuring algorithms.
- The emphasis is on the design of efficient algorithm

Criteria for analyzing Algorithms

Criterion for Analyzing Algorithms

We will measure algorithms in terms of the amount of computational resources that the algorithm requires:

- running time
- memory

Model of Computation

Model of Computation

- We want the analysis to be as independent as possible of the variations in machine, operating system, compiler, or programming language.
- Unlike programs, algorithms are to be understood primarily by people and not machines.

Model of Computation

Model of Computation

- Thus gives us quite a bit of flexibility in how we present our algorithms, and many low-level details may be omitted

Model of Computation

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- We call this model a *random access machine* or *RAM*.

Random Access Machine

Random Access Machine

- A RAM is an idealized machine with an infinitely large random-access memory.
- Instructions are executed one-by-one (there is no parallelism).
- Each instruction involves performing some basic operation on two values in the machine's memory